

Task Manager Application with Unit Testing

This project is based on **BCA-CC-605 Final Year Project**. The application is a Python-based Task Manager that supports task handling, JSON storage, CLI operations, and unit testing.

Step 1: Create the Task Class

- Create a Task class in task_manager.py.
- The class stores title, description, due_date, status, and priority.
- Validate missing fields during task creation.
- Validate invalid status and invalid priority values.

Allowed Status Values

- pending
- in-progress
- completed

Allowed Priority Values

- low
- medium
- high

Step 2: Create the TaskManager Class

- Add tasks
- Delete tasks by title
- Update task status or priority
- View all tasks
- Filter tasks by status or priority
- Save tasks to JSON file
- Load tasks from JSON file

Step 3: Add Error Handling

- Raise TaskManagerError when a required field is missing.
- Raise error when task title is not found.
- Raise error when status is invalid.
- Raise error when priority is invalid.
- Raise error when JSON file is corrupted.

Step 4: Create the CLI

- Run the command: `python cli.py`
- CLI allows adding tasks.
- CLI allows viewing tasks.
- CLI allows updating tasks.
- CLI allows deleting tasks.
- CLI allows saving tasks and exiting.

Step 5: Run Unit Tests

- Run the command: `python -m unittest discover -s tests`
 - Verify tasks are added correctly.
 - Verify tasks are deleted correctly.
 - Verify tasks are updated correctly.
 - Verify save and load JSON operations.
 - Verify handling of missing details.
 - Verify corrupted JSON handling.
 - Verify persistence after restarting application.
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Important Commands

```
# Example command to run CLI  
python cli.py
```

```
# Example command to run tests  
python -m unittest discover -s tests
```